



New York University
Computer Science Department
Introduction to Software Development
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Homework 6: *Practicing control statements and Strings.*

Deadline: *See NYUclasses for the deadline, 5% off per day after the deadline (3 days maximum).*

- Make an archive (zip file or compressed file) with all the java files (the .java files NOT the .class files) and post it on NYU Classes (StudentName_Homework_X.zip)
- It is your responsibility to make sure that the Zip files have your actual files. You may send the file to yourself by email to double check before you upload on NYU classes.
- If the graders cannot open the file, you will receive a grade of zero.
- If you send the .class files instead of the .java files (source files) you will receive a zero.
- Cheating will be severely addressed with an immediate zero on the homework and a report to the academic advisor and the administration.
- You will automatically lose 50% of the points for an exercise if the program does not compile and run correctly

Problem statement:

- You need to shift all elements of a 2D array to the right or to the left based on a flag entry.
- You need to solve this problem using **two different approaches**.
 - **Approach1:** create a method called arrShift1 which accepts two parameters as an input, a 2D integer array and an int call it flag and returns a 2D int array. The flag could be a positive int or a negative int.
 - If the flag is a positive int, shift all columns of the array to the right by the same number as indicated by the flag.
 - Inside this method you will extract each row of the input array and rotate it by one for n times where n is the flag int passed to the method.
 - Construct a new 2D array call it shiftedArr that holds the new shifted rows then return shiftedArr
 - **Approach2:** create a method called arrShift2 which accepts two parameters as an input, a 2D integer array and an int call it flag and returns a 2D int array. The flag could be a positive int or a negative int.
 - If the flag is a positive int, shift all columns of the array to the right by the same number as indicated by the flag.
 - Inside this method you will loop thru the 2D array once (only once) via nested for loops. While looping thru the array elements, you will shift the elements by n positions to the left or to the right based on the flag sign and where n == the flag passed in the method signature.

- Assume an Array, arr =

9	75	52	73	44
60	71	69	37	47
40	12	44	29	5
53	25	67	44	38
66	29	49	33	50
26	10	47	20	40
33	58	25	15	40
22	71	21	27	32
-
- If the flag is a positive int, shift all columns of the array to the right by the same number as indicated by the flag.

Example:

arrShift (arr, 2) will result in shifting all columns to the right by two positions as follows:

73	44	9	75	52
37	47	60	71	69
29	5	40	12	44
44	38	53	25	67
33	50	66	29	49
20	40	26	10	47
15	40	33	58	25
27	32	22	71	21

arrShift (arr, -1) will result in shifting all columns of the original array, arr, to the left by one position as follows:

75	52	73	44	9
71	69	37	47	60
12	44	29	5	40
25	67	44	38	53
29	49	33	50	66
10	47	20	40	26
58	25	15	40	33
71	21	27	32	22

- Assumptions:
 - the absolute value of the flag, f, where $0 < f \leq \text{array.length}$
 - The array is given as a part of the main method, you don't have to ask the user to input the array. i.e. don't use scanner.
 - Use the code template, below, as a template to insert your logic in each of the methods accordingly.

- Submittal:

Submit the java code, not the class file. use the main method as provided in the code sample below. Your code should be inside ArrShift1, ArrShift1, shiftRightbyOne, shiftLeftbyOne methods, and any additional method you want to add
You can/should use temp variables inside the methods as you see necessarily.

```

1 import java.util.Arrays;
2 public class Assignment6Driver {
3     public static void main(String args[]) {
4         int arr[][] = {{ 1, 2, 3, 4, 5 },{1,2,3,4,5}}; // test your code with other 2D arrays
5         int flag = 2; //test your code with other flag values; positive and negative
6
7         System.out.println( "Original Array via approach1: " + Arrays.deepToString(ArrShift1(arr,0)));
8         System.out.println("Shfted Array via approach 1: "+Arrays.deepToString(ArrShift1(arr,flag))+'\n');
9
10        System.out.println( "Original Array via approach 2"+ Arrays.deepToString(ArrShift2(arr,0)));
11        System.out.println("Shfted Array via approach 2: "+Arrays.deepToString(ArrShift2(arr,flag)));
12    }
13    //-----//
14
15    public static int[][] ArrShift1(int[][] A, int flag){
16        /*** Approach1      *** //
17
18        return A; // replace that with the shiftedArr
19    }
20    public static int[] shiftRightbyOne(int[] B){
21        // use this method with ArrShift1 method
22        return B;
23    }
24    public static int[] shiftLeftbyOne(int[] B){
25        // use this method with ArrShift1 method
26
27        return B;
28    }
29
30    /*** Approach2      *** //
31    public static int[][] ArrShift2(int[][] A, int flag){
32        //make sure you loop thru the arrayA once
33
34        return A; // replace that with the shiftedArr
35    }
36
37    //-----//
38
39 }

```